2 INTELLIGENT LOGISTICS

With the coronavirus pandemic causing a lockdown, our reliance on doorstep delivery has surged and, naturally, companies that are delivering essential items are now facing major stress on their supply chain. Technology can be a great enabler to navigate through these tough times. It ensures that deliveries reach us, while we work safely from our homes.

The customer can choose to get the expected time of arrival (ETA) alerts. This will ensure that the customer is prepared to collect the parcel with the already existing tracking link. Sharing driver information, digital transactions instead of cash-on-delivery, and one-time password (OTP) verification at the time of delivery, adds an additional layer of security to our packages.

Businesses should be ready to deal with uncertainties now, more than ever. Think of the worst case, best case and conservative case scenarios and arrive at the possible impact on your logistics for each use case—fleet required, number of drivers to be deployed, impact on fuel cost, warehousing needs, inventory management, etc.

Locus is the advanced supply chain decision-making engine that reduces costs, achieves order prioritisation, and enhances customer experience. Its geocoding engine converts the most ambiguous addresses into precise geographical coordinates with location tagging and geospatial-heat maps. Addresses are mapped correctly irrespective of the quality of address, leaving no room for error, thus avoiding misroute. It automatically directs your fleet/driver to use alternative and safer routes.

Locus helps to keep track on-ground resources and get instant alerts in case of any deviation from the planned route, exceeding break times, undesignated halts, dynamic alerts for maximum and minimum temperature, etc. Smart alerts allow the rider to mark any order completion to ensure optimal utilisation of resources, and thus reduce costs. It improves customer experience by informing them beforehand about potential delays with predictive alerts.

Live tracking provides delivery details and allows the dashboard admin to take a service level agreement (SLA) to breach corrective measures with transparency. It provides details like the current latitude and longitude, estimated time of arrival, contact details of customer care and the driver, resulting in enhanced customer experience.

Enroute Analytics allow comparing planned vs executed delivery routes. This helps to observe the driver's riding behaviour and analyse his compliance with the given plan. It lets you understand the rider's deviations from the plan to eradicate frauds.

Locus Dispatcher is an advanced and powerful route optimisation software. It does artificial intelligence (AI) backed route planning that has a negligible dependency on human intelligence. It supports on-demand dispatch planning and optimal handling of orders along with scheduled routing on the go. It provides automated recommendations on the best-suited vehicles based on traffic, volume, shipment, vehicle type, delivery times, stop durations, and more.

Locus software has a detailed, easy-to-use UI that is available for both iOS and Android mobile users. It provides electronic proof of delivery such as customer signature, product image, or QR code at the point of sale to authenticate your deliveries. It enables easy rescheduling, cancelling of orders, or a part of the order for enhanced customer experience and reduced logistics costs.

—Deepshikha Shukla

3 3D BIOPRINTING FOR ORGAN TRANSPLANTATION

The medical world has been able to find remedies for most issues that were once life-threatening for mankind in the past century and some of it, if not all, could be truly possible due to rapid advancements in technology. Since the first-ever successful organ transplant, great improvements have been made to increase the success rate of transplants. But the problem is that organ supply is not enough to suffice the demand, causing deaths from vital organ failure. 3D printing is being seen as the solution to this issue by researchers across the globe.

Next Big Innovation Labs, a Bengaluru startup founded in 2016, is working to develop innovative products and services for research and development and clinical applications in this field. The team was awarded the Idea2POC grant as well as the Biotech Ignition grant under the BIRAC scheme in 2017. The founders initially began making maxillofacial 3D models for doctors with the help of patient-specific scans.

After realising the need for customisable implants, the team started